

# NITISH KUMAR

## APPLIED AI ENGINEER

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Noida, India

## SUMMARY

Applied AI Engineer with 4 years of software engineering experience across Generative AI, LLM-powered applications, Python backend systems, and SAP S/4HANA. Hands-on experience building enterprise RAG and agentic AI workflows, LLM orchestration, APIs, asynchronous services, and reusable AI components. Experienced in AI evaluation, observability, traceability, validation, citation-aware responses, authentication, and reliability controls including checkpointing, idempotency, retries, and failure recovery. Currently contributing to an AI-assisted SAP custom-code assessment and remediation platform in pilot with a European aerospace client.

## EXPERIENCE

### Sopra Steria India

Jul. 2022 – Present

Senior Software Engineer – Applied AI & Backend Engineering

Noida, India

- **Enterprise GenAI platform:** Develop an AI-assisted SAP custom-code assessment and remediation platform using Python, FastAPI, LangGraph/LangChain, Celery, PostgreSQL, and Google Vertex AI for enterprise migration workflows.
- **RAG and agentic AI:** Built LLM-powered RAG and agentic workflows using embeddings, metadata filtering, semantic/hybrid retrieval, reranking, tool calling, prompt engineering, and citation-aware responses to improve retrieval quality and source attribution.
- **Durable AI orchestration:** Designed dependency-aware asynchronous workflows with checkpoints, idempotency keys, retries, resumable execution, and failure recovery across assessment, remediation, validation, and transport stages.
- **Trustworthy AI and guardrails:** Implemented validation, structural checks, diff controls, review gates, authorization boundaries, and controlled workflow transitions to reduce generated-code regression risk and improve transparency and reliability of AI-assisted changes.
- **Evaluation and LLMops:** Evaluated model and provider options across response quality, latency, reliability, token usage, and request-level cost to support model-routing and architecture decisions.
- **Observability and traceability:** Added logging, tracing, token tracking, latency monitoring, error monitoring, and execution visibility to troubleshoot AI workflows and continuously improve system behaviour.
- **Governance and security:** Built role-based workflows for administrators, project managers, and developers using JWT authorization, approval gates, scoped actions, and controlled lifecycle transitions.
- **Enterprise integrations:** Integrated SAP systems through PyRFC for custom-code extraction and migration workflows spanning SAP ECC and S/4HANA environments.
- **Engineering delivery:** Collaborated with engineering teams and stakeholders to translate requirements into technical workflows, identify implementation gaps, document system behaviour, troubleshoot issues, and improve production readiness.

## PROJECTS

### GPT Craft – Full-Stack Agentic LLM Application

FastAPI, React, LangChain, OpenAI GPT, MCP

- Built a full-stack LLM application for streaming conversational workflows with prompt pipelines, session context management, and short- and long-term memory.
- Integrated Model Context Protocol (MCP) for modular tool access, structured AI responses, tool calling, and reusable agent capabilities within LangChain workflows.
- Implemented authentication, asynchronous request handling, rate limiting, and request logging to improve security, scalability, and backend reliability.
- Instrumented token usage, latency, failures, and model behaviour to support debugging, evaluation, operational observability, and continuous improvement.

## EDUCATION

### Dr. A.P.J. Abdul Kalam Technical University (AKTU)

2018 – Jul. 2022

B.Tech in Computer Science

Lucknow, India

## SKILLS

**Generative AI:** LLMs, Generative AI, RAG, Agentic AI, Prompt Engineering, Context Management, Tool Calling, MCP, LangChain, LangGraph

**Retrieval & Models:** OpenAI GPT, Claude, Gemini, Google Vertex AI, Embeddings, Hybrid Search, Semantic Search, Reranking, FAISS, Pinecone, Citation-Aware Retrieval

**AI Reliability & LLMops:** AI Evaluation, Validation, AI Guardrails, Source Attribution, Traceability, Observability, Token Tracking, Latency Monitoring, Failure Recovery, Checkpointing, Idempotency, Retry/Recovery

**Programming & Backend:** Python, SQL, FastAPI, REST APIs, Async APIs, PostgreSQL, Celery, Redis

**Engineering & Delivery:** API Design, CI/CD, Distributed Workflows, Dependency Scheduling, Authentication, OAuth 2.0, JWT, Docker, Technical Documentation

**Cloud & Enterprise:** GCP, Google Vertex AI, AWS, SAP S/4HANA, SAP ECC, ABAP, PyRFC, WRICEF, DDIC